A Global Archive of River Gauges for Large-Scale Hydrologic Exploration

Abstract

1. Introduction

The systematic measurement of river stage and discharge is essential to having a baseline understanding of hydrological norms and extremes. They also provide the critical information necessary for simulation models to be compared, calibrated, and improved. These models, both local, national, and global serve as the foundation for flood forecasting and early warning systems, drought monitoring, water supply allocation, infrastructure design, hydropower operations, and environmental flow assessment. Without reliable gauge records, the accuracy and credibility of both operational forecasts and scientific analyses of river behavior are fundamentally limited.

Despite this importance, the global distribution of stream gauges is uneven, and the number of stations actively contributing to archives has been declining since the 1980s (Ruhi et al., 2018). Established in 1988, the Global Runoff Data Centre (GRDC) is the most prominent effort to compile river discharge observations at the global scale (GRDC, 2024). The GRDC currently holds daily and monthly discharge records from more than 9,800 stations across 160 countries, representing approximately 435,000 station-years of data with an average record length of 44 years. However, even the GRDC's impressive data repository has only about 14% of stations remaining active as of 2020, and coverage is heavily skewed toward North America and Western Europe. While these regions maintain dense, well-funded monitoring networks, many developing regions, often those most vulnerable to floods, droughts, and water scarcity, have inadequate coverage (Mishra and Coulibaly, 2009). Furthermore, data-sharing barriers related to national policy, transboundary disputes, and institutional fragmentation mean that even where gauges exist, their records may not be accessible to the broader scientific community or to global modeling efforts.

These gaps pose a challenge for large-scale hydrologic models such as the GEOGLOWS ECMWF River Forecast System, whose accuracy depends on observed data for bias correction, calibration, and validation (Hales et al., 2022; Sanchez Lozano et al., 2021). Building a comprehensive, well-documented global gauge dataset is needed for not only improving global streamflow predictions, but also for advancing our collective ability to manage water resources, reduce disaster risk, and improve water security in data-scarce regions worldwide.

The two purposes of this paper are first to describe a dataset of in situ river stage and discharge measurements compiled for global hydrology modeling, and second to analyze the data for its strengths and weaknesses and correlations with socioeconomic indicators. The gauge measurements themselves are not all publicly available, so we do not publish the full dataset for download in connection with this article. We instead provide a collection of scripts and documentation to retrieve the data described separately. We describe the dataset in terms of its various summary properties including its spatial coverage in terms of political boundaries as well as watersheds, and also the temporal availability of the measurements. The detailed descriptors provide a basis for understanding the dataset without needing to distribute the entire catalog of measurements.

2. Building the dataset

2.1. Data providers

Describe who gave us the data. National agencies? International groups? What kind of national agency is it (hydrology group, met group, emergency management group)? How did we find them>

Do we want to take about water level data vs flow data at all?

Our data sources include official country-specific APIs, national web portals, collaborative international platforms/initiative (such as GRDC and ArcticNet), and exclusive datasets provided for us for specific research projects. The following table details the providers from which data was collected as well as the approximate spatial coverage of the dataset.

2.2. Collecting data

Data formats, different APIs, this could also be where we talk about barriers in the collection process - Could answer the part of the thesis where we talk instructions for how someone could repeat our processes?

Table 1: Data Providers and Spatial Coverage

Name	**Coverage**
ANA (Agência Nacional de Águas)	Brazil
Bangladesh	Bangladesh
Belize	Belize
Bhutan	Bhutan
BoM (Bureau of Meteorology)	Australia
Caravan (derived from CAMELS)	Global / Multi-country dataset
CEDEX (Centro de Estudios y Experimentación de Obras Públicas)	Spain
DHM (Department of Hydrology and Meteorology)	Nepal
EA (Environment Agency)	Europe
Eaufrance (public water information service)	France
ENEE (Empresa Nacional de Energía Eléctrica)	Honduras
Ethiopia	Ethiopia
GRDC (Global Runoff Data Centre)	Global
HYDAT (Hydrometric Data)	Canada
IDEAM (Instituto de Hidrología, Meteorología y Estudios Ambientales)	Colombia
IMHPA (Instituto de Meteorología e Hidrología de Panamá)	Panama
INAMEH (Instituto Nacional de Meteorología e Hidrología)	Venezuela
INAMHI (Instituto Nacional de Meteorología e Hidrología)	Ecuador
India-WRIS (Water Resources Information System)	India
INDRHI (Instituto Nacional de Recursos Hidráulicos)	Dominican Republic
Israel Meteorological Service	Israel
Ivory Coast	Ivory Coast
Kenya	Kenya
Malawi	Malawi
Paraguay	Paraguay
SENAMHI (Servicio Nacional de Meteorología e Hidrología)	Peru
SERVIR Mekong	Mekong Region (Southeast Asia)
SHI (State Hydrological Institute)	Russia
SIH (Sistema de Información Hidrológica)	Mexico
Somalia	Somalia
Tanzania	Tanzania
USGS (U.S. Geological Survey)	USA
RWB (Rwanda Water Resources Board)	Rwanda
DGRE (Direction Générale des Ressources en Eau)	Togo

We saw two types of organizational structures of data management. The first data-organization style had all associated metadata available in the individual timeseries files. These were typically obtained through API queries. The second data-organization style involved two separate data streams. The first data stream were the files containing time series measurements for each gauge, and the second was a single file containing metadata for all stations. Real-time sub-daily data is generally accessible through API interfaces rather than static downloads.

Despite source diversity, core metadata elements include gauge identification number, station name, river name, geographic coordinates (latitude and longitude), and time series measurements. Supplementary metadata varies like administrative divisions of country, collecting organizations within country, station elevation, hydrological classification, municipality information, station start dates, API end points, quality indicators, and distance to outlet.

For further analysis on all gauges, we standardized the metadata. For stations lacking elevation data, we derived values by averaging NASADEM and ALOS DSM measurements. Missing administrative divisions were populated through reverse geocoding via geonames.org. Climate classifications which were not present in any of the metadata and were filled using Koppen climate classifications. They were extracted from published raster datasets based on present conditions [cite that nature paper]. Portions of the gauge were provided with GEOGLOWS V2 river IDs. We further calculated analytical metadata including temporal availability, observation completeness). Gauges were designated as "active" if their last available date matches the date of extraction, meaning that we assumed they are still active.

For quality assurance processes, we standardized the missing data across all sources using a uniform identifier (NaN). This standardization presented significant challenges due to the heterogeneous representation of missing values among different and same sources including "0", "-", "-9999", and simply blank. This identification proved particularly complex in river on arid regions or small Strahler order, where legitimate zero flow measurements could be misinterpreted as missing data. Also, in same sources multiple missing data indicators within single datasets were found to be used like "0" and "-". Stations with missing geographic coordinates required additional research either through source web portals if available or from literature review on published journal.

2.3. Organizing measurements

Do we still want to publish the metadata table? If so, this is where we could describe it. Data formats we store it in and why, how up to date things are

Data is stored on an AWS S3 bucket, divided into two main sections: sources, and production data. For every gauge, it is partitioned into a group matching the country it came from and the data provider. The sources contain the raw data files that we collected. For our production section, we have three primary items for each country-provider group: a catalog.csv, sites.gpkg, and a measurements directory.

The catalog csv has these columns: gauge_id, latitude, longitude, river_name, final_river_id, discharge, and water_level . A description of each follows...

The sites.gpkg is simply the catalog csv converted to a spatial format. This format allows for easy geopspatial analysis.

The measurements directory contains one CSV for each gauge, containing 2-3 columns: datetime, discharge (optional), and water_level (optional)... Description....

A master catalog was created to anaylze the total metadata of all the gauges. This catalog contains additional information collected by previous graduate and post-doc students, and here is a description....

2.4. Disseminating / Barriers to access

The primary barrier to global river gauge data access lies not in its data availability but in the fragmented nature of information about these resources. Numerous countries provide unrestricted hydrological observations, yet awareness of those information remains limited. For instance, Spain and Mexico offer readily downloadable datasets without API requirements, but the hosting platforms lack visibility and not many know about its existence. A secondary barrier involves the technical expertise required to understand API endpoints and programming. While manual downloads are enough for isolated stations to study small areas, programmatic approaches become essential when managing thousands of continuously updating gauges across the country. To address these challenges, we have developed country-specific GitHub repositories containing code examples and documentation to facilitate streamlined data access. (???)

3. Analysis of the collected data

3.1. Geographic Analysis – Where are the data?

The dataset consists of over 37,000 gauges. The dataset contains data from 135 countries based on the World Bank country boundaries. We analyzed the geographic location of the gauges and present an analysis to summarize where the data are primarily available and where they are missing. The figure below shows the location of 36,000 of the gauges. These are gauges that are publicly available and therefore can be published without violating privacy issues. The map marks which stations for discharge versus stage data.

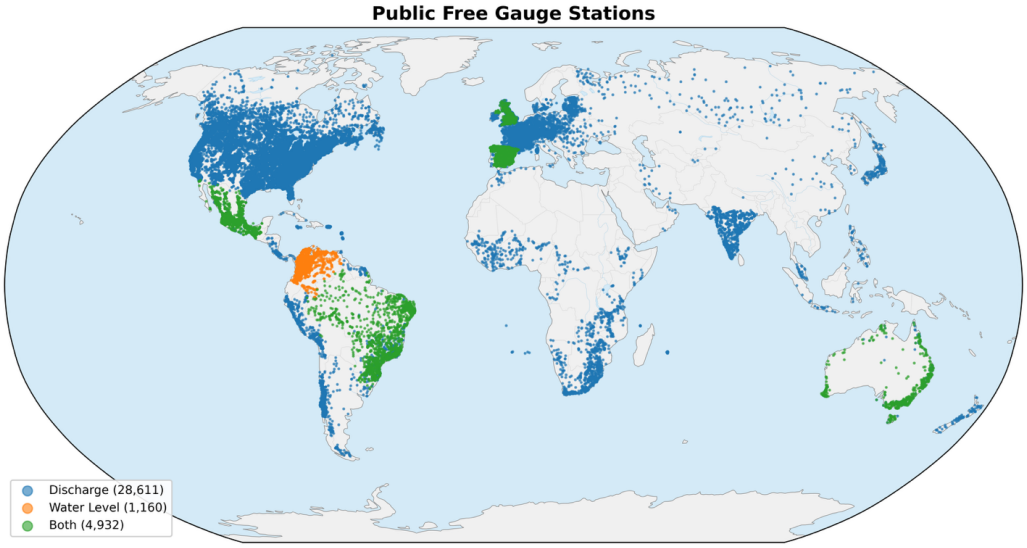


Figure 1: Locations of publicly available gauges, marked by whether they report discharge, water level, or both.

Examining the figure shows some interesting patterns in spatial biases around the world. This dataset does not have much data in the Nile River Basin. While we know data exists in that area, data sharing protocols prevent it from being included in global datasets. We also note that as expected there are very few gauges in areas that are primarily desert including in the Sahara Desert, the Arabian desert, Patagonia, and the Australian desert. We also note from this figure that certain areas of the world, such as Europe, the United States and Canada, are much more densely covered than other areas.

The United States contains over 11,000 gauges making up about a third of the dataset.

In the dataset, there are 129 countries without any gauge data. However, many of the countries are smaller islands such as Maldives, a land locked but small country such as Vatican City, or a small territory owned by another country such as Guam. The following figure shows the area size of countries. Many of the countries are smaller than the United States state of Rhode Island which is a little smaller than 3000 square kilometers. Therefore, while the number of countries missing gauges is high, the amount of area is much smaller.

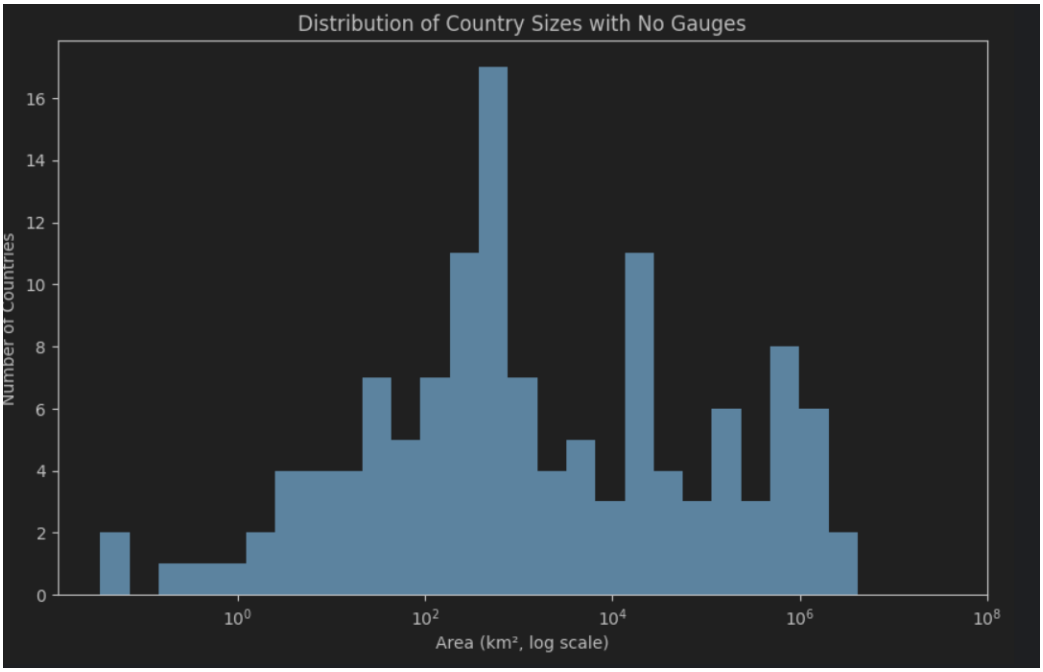


Figure 2: Distribution of the areas of countries with no gauges.

We examined to see if the countries with data were biased at all based on income levels. To assess this, we compared the number of gauges in each country with the country's Gross National Income (GNI) per capita, as reported by the World Bank. We selected 2019 for the analysis because it provided the most complete and recent dataset available. Using an earlier year also ensured that any potential relationship between GNI and gauge distribution could be more clearly identified, given that our dataset does

not include data from the most recent years. Three countries—Venezuela, Monaco, and Liechtenstein—contained gauges in our dataset but lacked GNI values, and therefore their gauges were excluded from classification. We also used information on the size of the countries and populations based on World Bank data. Gauge locations were assigned to countries using the World Bank's country shapefile, which was selected to ensure consistency with the GNI and population datasets.

After obtaining the GNI, we used it to classify countries as low income, lower middle income, upper middle income and high income in accordance with the World Bank's categories for these categories. This allowed us to see how the gauges were spread across higher and lower income areas.

Table 2: Gross National Income (GNI) split into 4 categories in accordance with World Bank requirements.

Low Income	GNI $<$ \$1035
Lower Middle Income	\$1036 $<$ GNI $<$ \$4085
Upper Middle Income	\$4086 $<$ GNI $<$ \$12615
High Income	GNI $>$ \$12615

We also examined each income group for how many people and area there were on average per gauge. In this analysis, the United States gauges were removed. This is because there are many gauges in the United States in our dataset that it had a significant impact on the averages of high-income category. Therefore, it was removed to give a more representative result.

Table 3: Average gauge counts, population, and area per gauge by income group, excluding the United States.

Income Group	Average gauges per country	Avg 1000 people per gauge	Avg 1000 km^2 per gauge	Number of countries with gauges
Low Income	13.23	1738.81	42.95	16
Lower Middle Income	35.51	1988.91	16.84	33
Upper Middle Income	103.83	432.94	23.23	34
High Income	232.42	55.08	5.18	37

The following map shows the countries that are missing data grouped by incomes based on the GNI. There are countries with higher income that have

no gauges in the dataset, but many of these countries have explanations for why no data may exist. Iceland is a smaller island and therefore does not have any major rivers. Finland and Norway similarly do not have many large rivers due to the coastlines, and both are north with lots of snow. The map shows that many of the countries missing data are in Northern Africa and Western Asia.

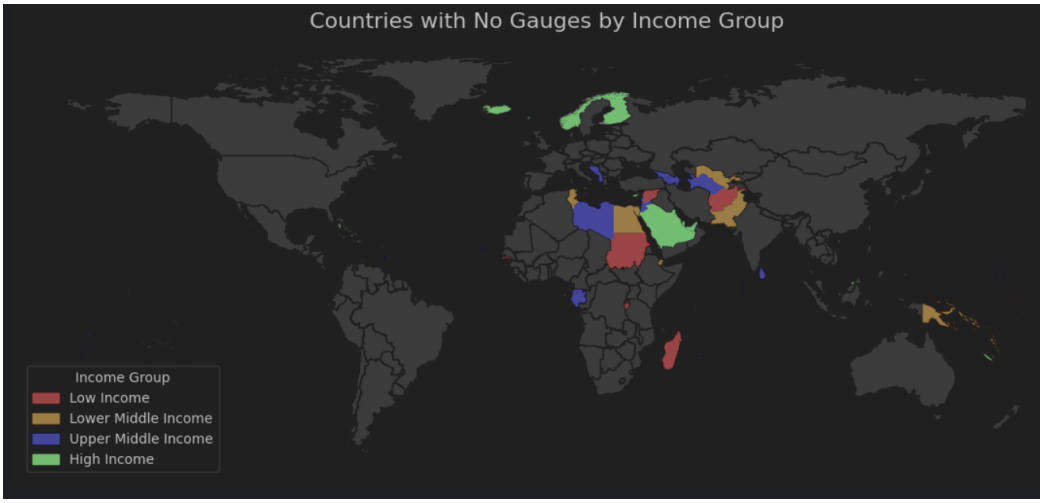


Figure 3: Countries with no gauges, grouped by income category based on GNI.

In addition to looking at which countries have data, we also observed which climate zones have more data available. We looked at the representation of gauges in the different Koppen Climate zones.

The largest portion of the gauges are in the temperate climate despite being the smallest percentage of land area. However, this is the climate with the greatest population of people. There are very few gauges in the polar regions, which could be due both to the lack of people living there as well as the lack of rivers to analyze.

The gauges in the dataset are matched with RFS streams. (*Describe in more depth?*) All gauges that have been matched used this global hydrology to show what areas are covered by a gauges. And area is covered by a gauge if there is any gauge downstream of that river. The following map displays the results of this analysis.

The map highlights

Slice and dice fifty ways.

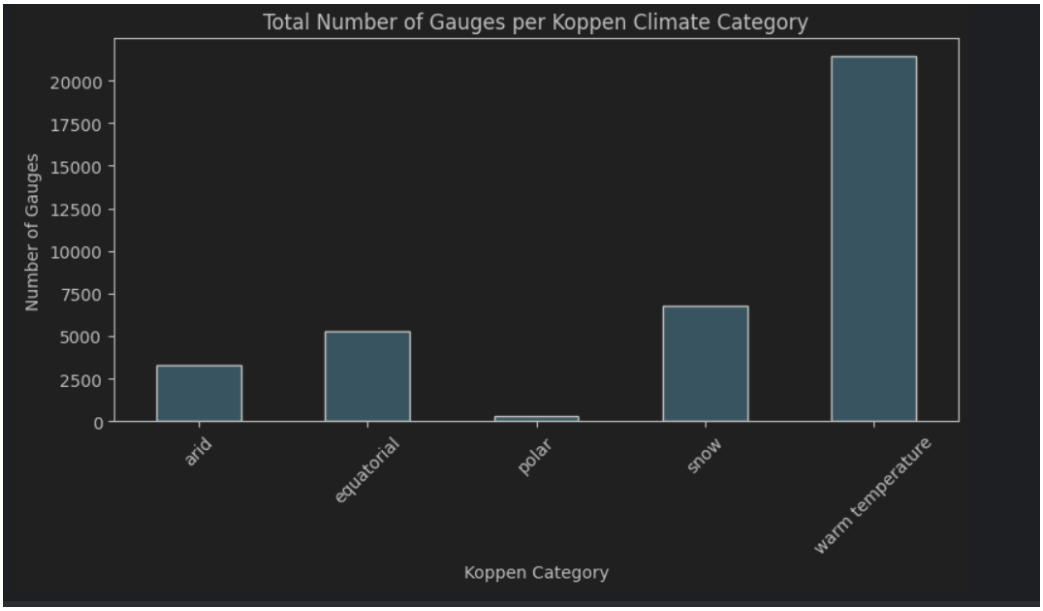


Figure 4: Total number of gauges per Koppen climate category.

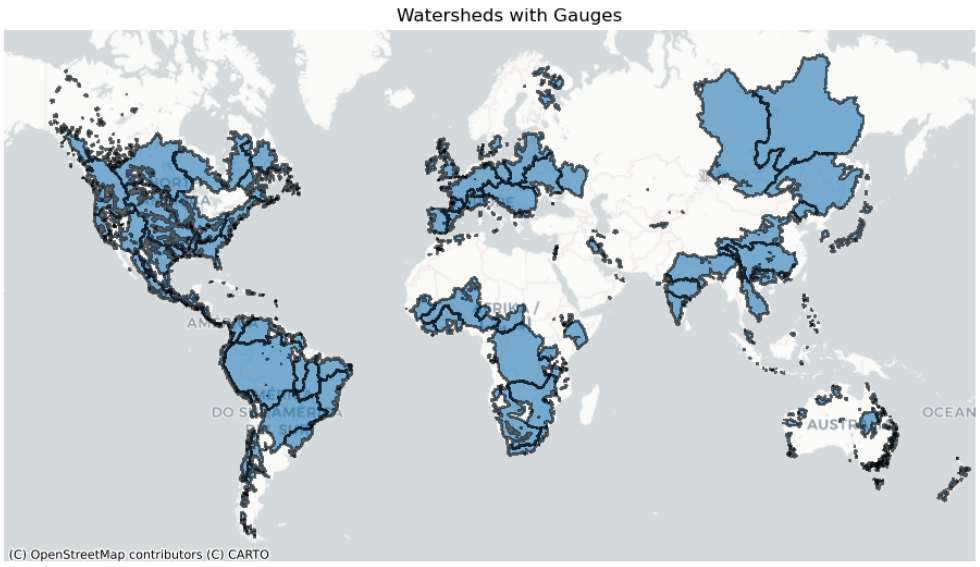


Figure 5: Watersheds with gauges.

Watersheds that have rivers

How many gages do we know about that are not in our database (e.g., China, Russia, Nile, the stans, Indus, Tigris, Euphrates, Paraná

ALASKA?

Places where we have no GeoGlows stream reach for a gage. (QUESTION: Who does have a model for that location? (e.g., does GloFAS or Flood Hub include that number of gages?)

Continent – world bank regions? Hard to find specific continent shapefiles

3.2. Temporal Analysis – When are the data?

Temporal resolution – how frequently are most of the measurements

This section presents temporal analyses of the globally distributed gauge dataset compiled in this research. As shown in Table 4, temporal coverage varies substantially across regions, reflecting historical differences in hydrological monitoring investment and institutional capacity. Europe and North America hold the deepest historical records, with gauges dating back to 1806 and 1862, respectively, and together account for over 95% of gauging stations predating 1930. Africa, South America, and Australia each contribute modest pre-1930 representation in tens of stations, while Asia, South Asia, and Central America contribute fewer than 10 gauges before that mark. The Middle East stands apart as the only region with no gauge records before 1983. Because pre-1930 coverage is concentrated almost entirely in Europe and North America, temporal analysis in this study begins in 1930, after which enough regions have established monitoring records to support meaningful cross-regional comparison. Further, record availability extends through the early-to-mid 2020s across most regions, with North America and Central America offering the most current data through 2025, and Asia and the Middle East trailing with records last updated through 2021.

Sub-daily measurement availability follows a similarly uneven pattern: Australia and South America lead with 63.2% and 52.1% of their gauges providing sub-daily data, while Africa, Asia, the Middle East, South Asia, and Central America report none. These disparities suggest that data-sparse regions not only lack spatial gauge density but also suffer from shallower temporal records and coarser measurement resolution, compounding the challenge of using observational data for global hydrological model calibration and validation in those areas.

Figure 6 shows the number of active gauges per region from 1930 to the latest gauge available year. North America and Europe dominate the

Table 4: Temporal coverage of gauge records by region.

Region	**Earliest gauge date**	**Latest gauge date**	**Number of Gauges before 1930**	**% Gauges with subdaily data**
Africa	1903	2024	41	0
Asia	1902	2021	7	0
Australia	1905	2022	18	63.2
Central America	1936	2025	0	0.2
Europe	1806	2024	707	26.3
Middle East	1983	2021	0	0
North America	1862	2025	1468	24
South America	1901	2022	44	52.1
South Asia	1913	2024	2	0

global gauge network throughout this period, with North America peaking at approximately 9,000 active gauges around 1990 and Europe reaching roughly 5,500 by the mid-2010s. The rise in the number of gauges globally around 1980 can be attributed to the WMO and UNESCO initiatives to expand global hydrological monitoring following the International Hydrological Decade (1965–1974) (More, 1980). Studies on dams reveal that the construction of dams peaked globally between the 1960s and 1980s (Perera et al., 2021), which could have caused a proportional increase in gauging stations.

Both regions exhibit a marked decline in active gauges after approximately 2015, suggesting either network contraction or reporting lags in recently compiled data. South America shows a notable expansion beginning around 1980, rising from near zero to approximately 1,500 active gauges, where it remains relatively stable through 2020. All other regions — Africa, Asia, Australia, Central America, the Middle East, and South Asia — maintain consistently low gauge counts throughout the entire period, remaining nearly indistinguishable from zero on this scale. The major decline in freshwater monitoring networks stems primarily from insufficient funding and shifting government priorities (Ruhi et al., 2018).

This persistent underrepresentation underscores the severe temporal and spatial data gaps that characterize the global gauge network outside of North America and Europe, with direct implications for the reliability of global

hydrological models in those regions.

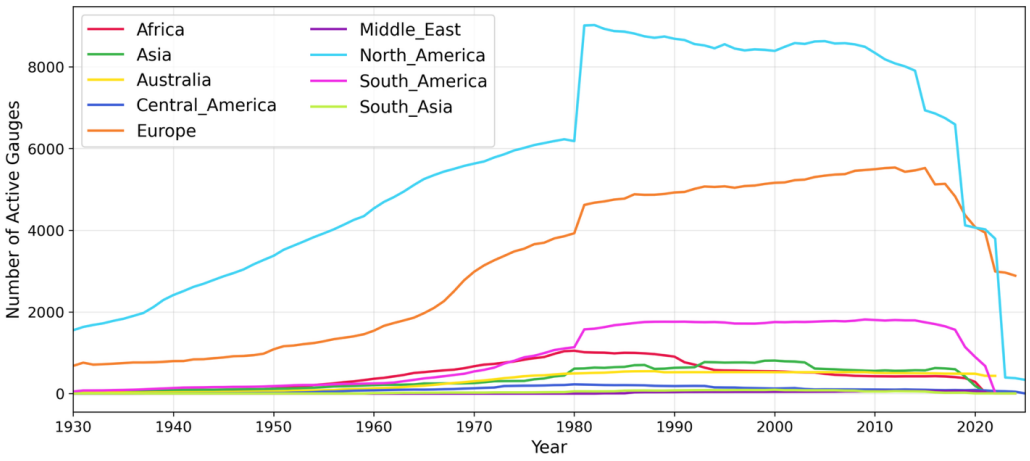


Figure 6: Number of active gauges per region from 1930 to the latest year with available data.

The decadal heatmap of active gauges (Figure 7) by region reveals distinct trajectories in global monitoring capacity over the past nine decades. North America reached peak coverage in the 1980s with 8,862 active gauges, followed by a gradual decline to 6,935 in the 2010s. Europe shows the most consistent growth trajectory, expanding from 747 gauges in the 1930s to 5,522 in the 2010s without interruption. Africa, by contrast, peaked in the 1980s at 999 gauges and has since declined to 424 in the 2010s, representing a more than 50% reduction in active monitoring capacity over four decades. South America grew steadily through the 1980s and has remained relatively stable since, while Central America, Australia, and Asia maintain a modest but measurable presence. The Middle East and South Asia remain severely underrepresented throughout the entire period, with the Middle East recording zero active gauges before the 1980s and South Asia never exceeding 85 gauges in any decade. Taken together, these patterns reveal that global gauge network expansion was largely a mid-twentieth century phenomenon concentrated in the Global North, with several regions now experiencing declining coverage rather than growth.

Figure 8 illustrates the annual number of gauges beginning and ending operation between 1930 and 2024. We assumed the gauges to be active at present if the gauges were active on the date of last measurement. From 1930 through the late 1970s, new gauge installations consistently outpaced

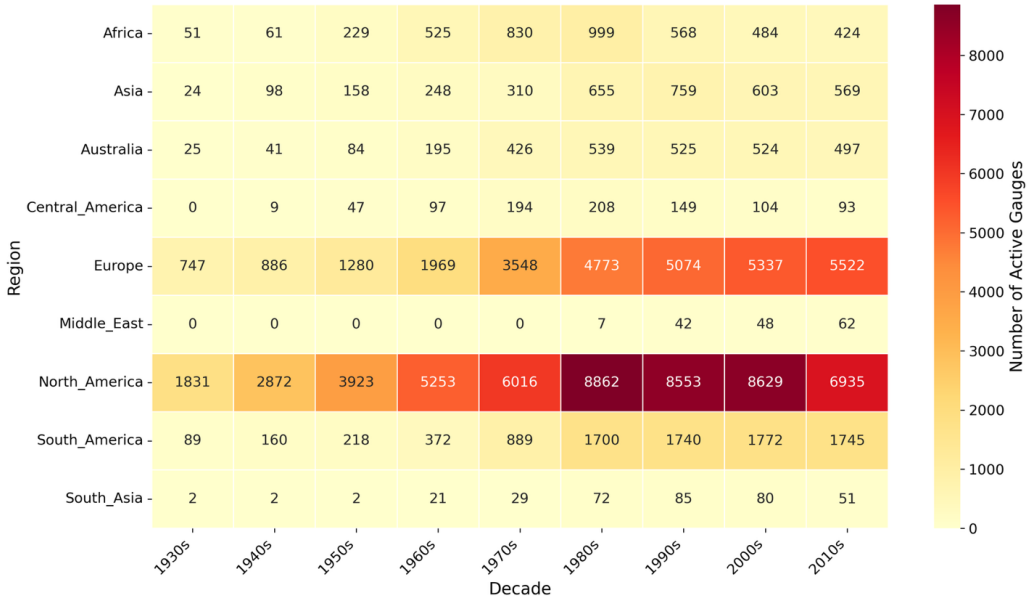


Figure 7: Decadal heatmap of active gauges by region.

closures, reflecting the global expansion of hydrological monitoring networks during the mid-twentieth century. A pronounced spike of approximately 4,100 gauge starts occurs around 1980, the origins of which warrant further investigation but may reflect a coordinated international monitoring initiative or a major data contribution from a single national agency during that period. Beginning in the mid-1980s, gauge endings become increasingly frequent, and the balance between new starts and closures shifts toward net loss. This decline accelerates markedly in the late 2010s, with a peak of nearly 3,000 gauge endings recorded around 2018–2019. While some portion of this recent decline may reflect genuine network contraction, it is likely also attributable to reporting lags in recently compiled data, where records for active gauges have not yet been updated through the present. Nonetheless, the overall trajectory is clear: global gauge network growth stalled and in many regions reversed after the mid-1980s, with implications that are particularly severe for data-sparse regions that never achieved adequate coverage to begin with.

Figure 9 shows the distribution of gauge record durations by region, expressed as a percentage of stations within each duration category. Europe and North America demonstrate the most favorable duration profiles, with

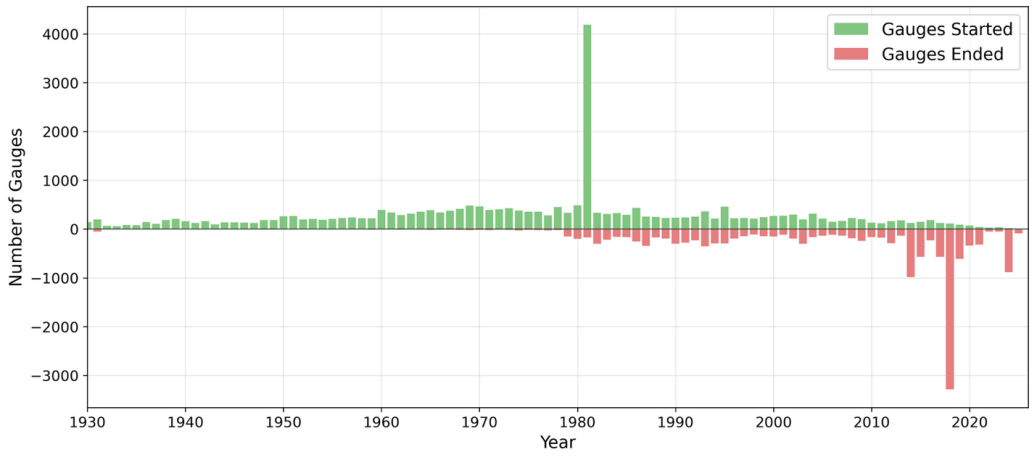


Figure 8: Annual number of gauges beginning and ending operation, 1930–2024.

large proportions of stations operating for 30 years or more — the threshold commonly recommended by the World Meteorological Organization for reliable hydrological analysis. North America notably contains the highest share of records exceeding 100 years, reflecting its long institutional history of systematic streamflow monitoring. In contrast, the Middle East is heavily weighted toward short-duration records of fewer than 10 years, consistent with its late network establishment in the 1980s. Asia exhibits a disproportionately large share of very short records of under five years, compounding the data quality concerns identified in the missing data analysis. Africa, despite having gauges dating to the early twentieth century, shows a substantial proportion of short-duration records, suggesting significant discontinuity and gauge turnover within its network over time. Australia and South America present more balanced duration profiles, with meaningful representation in the 20–50 year range. Taken together, these distributions indicate that outside of Europe and North America, a large fraction of the global gauge network does not meet minimum duration thresholds for robust hydrological modeling, further constraining the utility of available data in regions where observational coverage is already limited.

Some relevant research papers for this section

(Lanfear and Hirsch, 1999) talks about how USGS gauges are changing with time

(Ruhi et al., 2018) talks about decline in global gauge networks

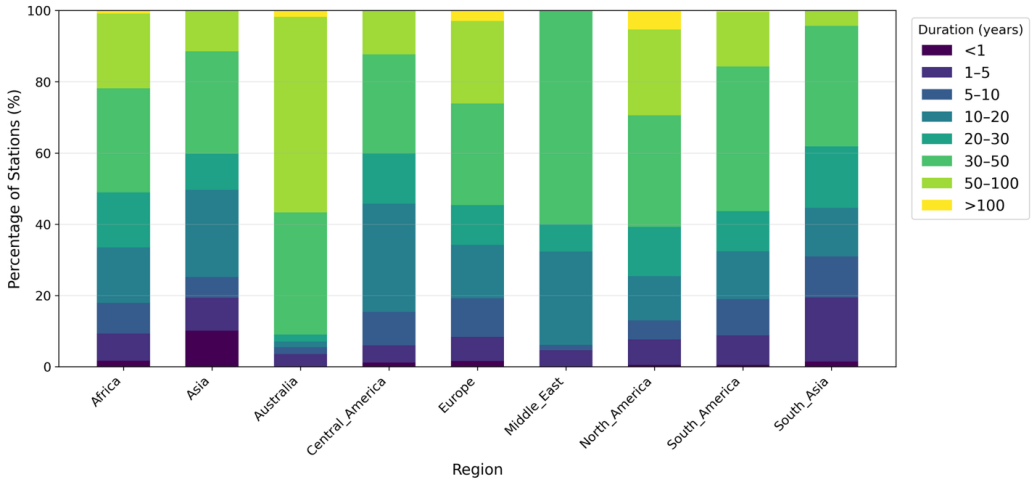


Figure 9: Distribution of gauge record durations by region, expressed as a percentage of stations within each duration category.

Figure 10 presents the distribution of missing data percentages across gauge records by region, offering insight into data quality independent of network size. Australia and the Middle East demonstrate the highest record completeness, with median missing data rates near zero, suggesting that where gauges exist in these regions, their records are generally well maintained. North America and Europe similarly show low median missingness, though both exhibit long upper whiskers extending to 100%, indicating that a subset of gauges within these otherwise data-rich regions have severely incomplete records. Asia shows the poorest typical data quality, with a median missing data rate of approximately 38% and a wide interquartile range, meaning that even where gauge records exist, substantial portions are often unavailable. Central America and Africa show intermediate but concerning levels of missingness, with medians near 30% and 15–20% respectively and numerous high-value outliers. Notably, the Middle East and South Asia — regions with the fewest gauges overall — do not exhibit the worst data quality, suggesting that network scarcity and record incompleteness are distinct but compounding challenges. Together, these findings indicate that the effective data available for hydrological modeling is considerably less than raw gauge counts alone would suggest.

Some relevant research papers for this section

(Lanfear and Hirsch, 1999) talks about how USGS gauges are changing

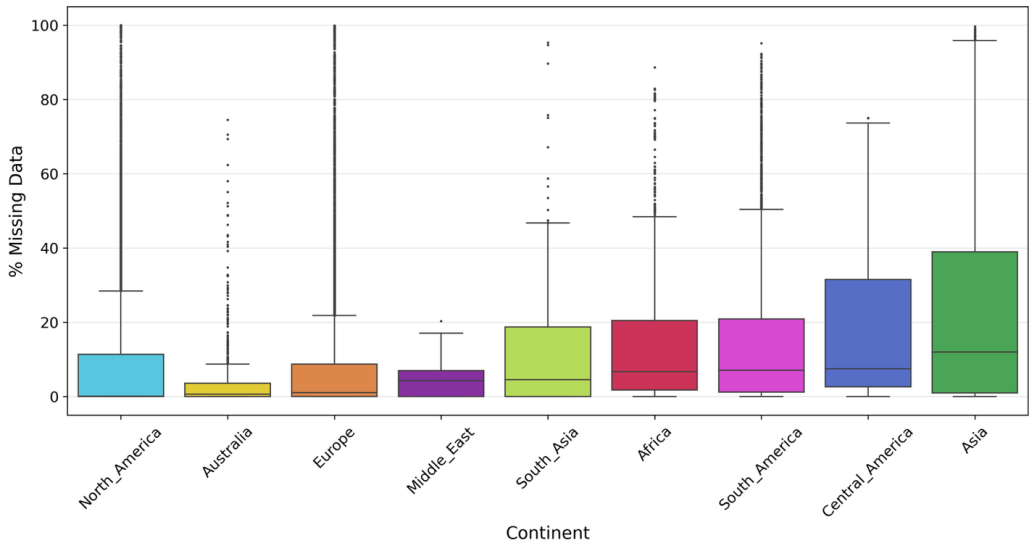


Figure 10: Distribution of missing data percentages across gauge records by region.

with time

(Ruhi et al., 2018) talks about decline in global gauge networks

4. Discussion

Analysis that is possible because of this broader dataset.
Things that have been accomplished because the data are where they are.

- Talk about RFS Bias Correction impact here AS AN EXAMPLE. NWM as an example.
- Show the Yubin's togo example showing that the global bias correction benefits from nearby gages... leading into...

Things that COULD be accomplished with this broader dataset (drawing on examples that already use large datasets

- Hydro SOS from WMO
- GloFAS (uses same forcing as GeoGlows... could expand)

Things that are significant impacts because of the gaps. Vulnerability/social issues/resilience.

Why the gaps.

How might they be resolved? (can we address that?)

5. Conclusion

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